From Anonymous Shapes to Named Places

A Tool for Braille and Place-Semantic Annotation of Tactile Maps

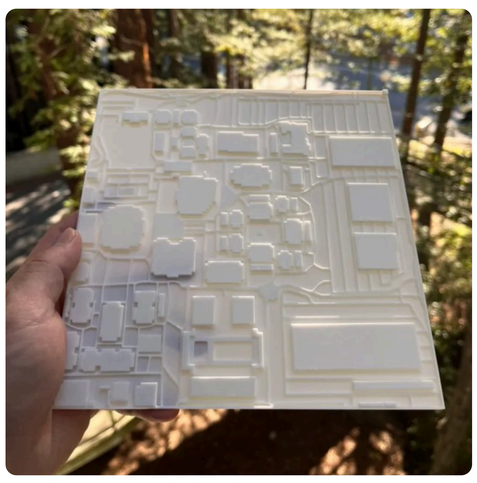
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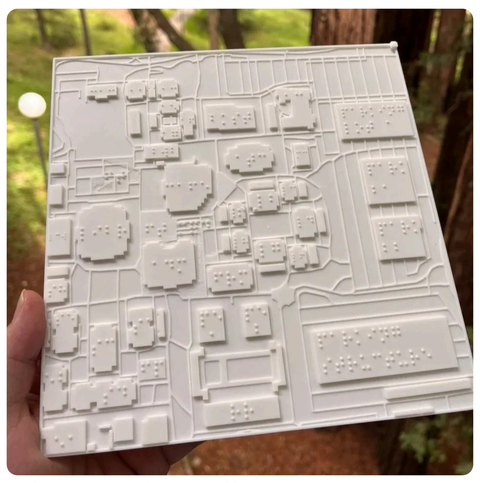
Try the tool

On a tactile map, a building unde

Our tool adds Braille labels to tactile maps that already e



Before · no labels



After · with Braille



Why we need it

A tactile map gives th

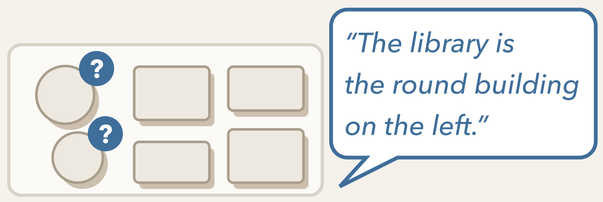
1



Felt, but not named

On a Touch Mapper† print, every building is a shape with no name.

2



Described, but hard to match

A spoken description is hard to tie to the right shape.

3

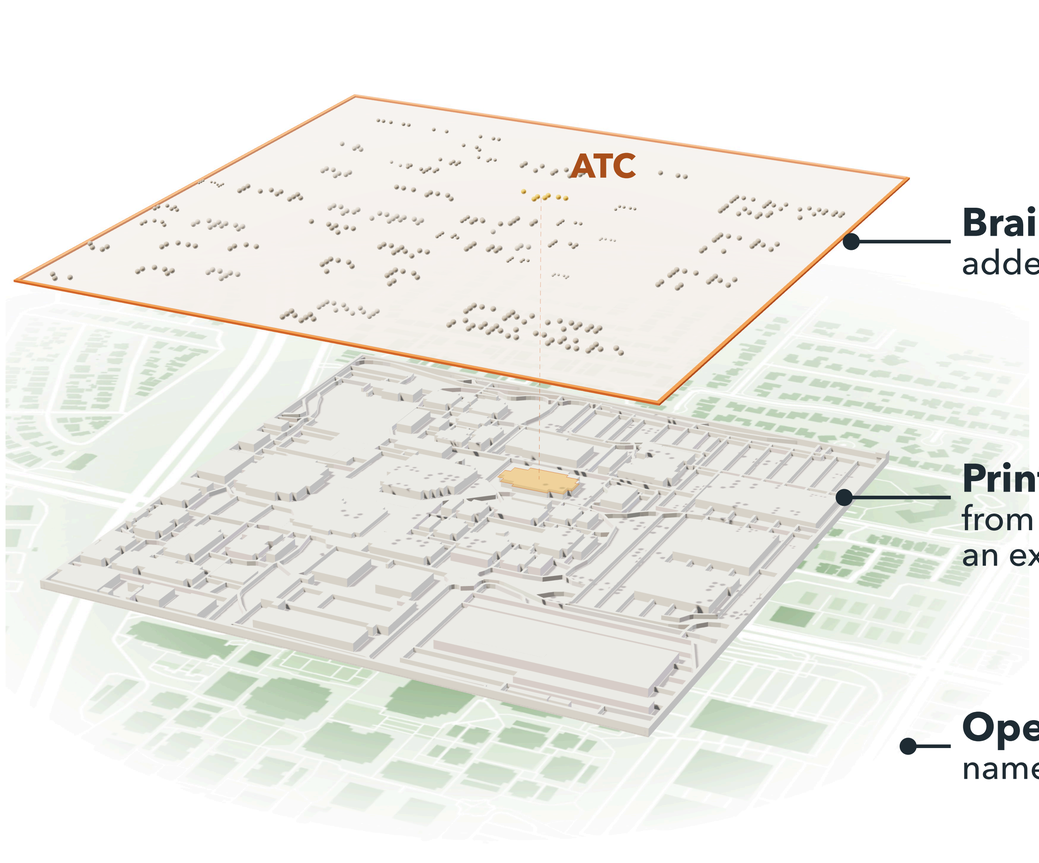


Hard to label by hand

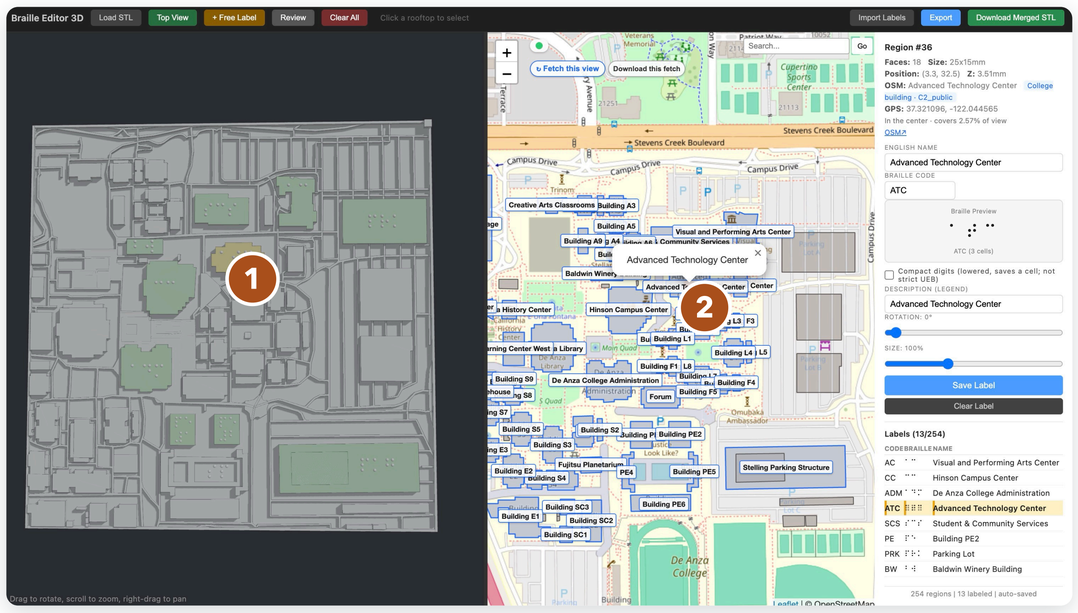
Placing each label, printing it safely and editing the 3D model take time and skill.

What our tool does

Braille becomes its own layer, linked to each place



3D model from Touch Mapper | **OpenStreetMap** | **Label**



1. Load the Touch Mapper model, click a building.
2. Click it on OpenStreetMap. Name and Braille fill in.
3. Review, then export. The dots are print-safe.

What it opens up

One label, many ways to read it

Today

- Braille on the print
- Braille legend
- Print-ready 3D model

Each label links

a shape on the print ↓ a place on OpenStreetMap

Future

- Audio readout
- Symbols beyond Braille
- AI agents
- Shared Braille map

Project page leolee7.github.io/#project-tactile-annotation · **Paper** doi.org/10.1145/3797867.3841235

†Touch Mapper, an existing tool that makes tactile maps from OpenStreetMap: touch-mapper.org

*Work unrelated to position at Amazon. Supported by AFOSR award FA9550-24-1-0149.

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